

MD. ROFAZ HASAN RAFIU

AI Systems Engineer • Software Architect

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PROFESSIONAL SUMMARY

Computer Science & Engineering student at RUET specializing in **Production AI Systems, Local Small Language Models (SLMs), In-Browser Computer Vision**, and scalable backend infrastructure. Experienced in bridging machine learning inference with resilient distributed infrastructure: local SLM diagnostics (Ollama/Qwen), client-side WebAssembly OpenCV, deterministic latency fallbacks, and multi-tenant PostgreSQL backends.

EDUCATION

- Rajshahi University of Engineering & Technology (RUET) — B.Sc. in Computer Science & Engineering**2023 – 2027
 - Admitted through national combined engineering entrance examination (CUET, KUET, RUET) ranking in top merit percentile.
 - Coursework: Data Structures & Algorithms, Object-Oriented Design, Database Systems, Computer Networks, Discrete Math.
- St. Joseph Higher Secondary School — Higher Secondary Certificate (HSC), Science**2022
 - GPA: 5.00 / 5.00

TECHNICAL SKILLS

- AI & Machine Learning:** Local SLM Inference (Ollama, Qwen 2.5, Llama 3.2), Structured Outputs (JSON Schema/Zod), OpenCV.js (WebAssembly), TensorFlow.js (BlazeFace), OCR (Tesseract.js), Deterministic Fallbacks, Prompt Engineering.
- Backend & Systems:** Python (FastAPI, Flask), TypeScript, Go (1.22+), Node.js (Next.js 15 App Router), RESTful APIs, WebSockets (Socket.io), System Design, Microservices, OWASP API Security, JWT, Role-Based Access Control (RBAC).
- Databases & Storage:** PostgreSQL (Neon, Aiven), Prisma ORM 6.10, Redis (Caching, Rate-Limiting), SQLite, Dexie.js (IndexedDB PWA Storage).
- Cloud, DevOps & Tools:** Docker, NGINX, GitHub Actions (CI/CD), Edge Caching (SWR, CDN Cache-Control), Linux/Bash Scripting, Git, Jest.

FEATURED ENGINEERING PROJECTS

- Digital School & Exam Platform (Rofaz Academy)** [Next.js 15, TypeScript, PostgreSQL, Prisma, Ollama, TensorFlow.js] **Live:** rofazacademy.dev
 - Local SLM Diagnostic Pipeline:** Engineered an automated assessment diagnosis engine integrating local SLMs (Qwen 2.5 3B / Llama 3.2 via Ollama) with temperature 0.3 structured prompting to extract root-cause conceptual mistakes from raw submission data.
 - Deterministic Fallback Reliability:** Architected a sub-10ms algorithmic fallback calculating topic-level error distributions and model explanations directly from PostgreSQL, ensuring 100% diagnostic reporting availability during local inference saturation.
 - Edge Scalability & Caching:** Configured edge CDN caching headers (Stale-While-Revalidate) and SWR synchronization on Next.js 15 routes, reducing origin database connection load by up to 70% during peak exam traffic.
 - Multi-Tenant Architecture:** Designed relational PostgreSQL schemas with Prisma ORM 6.10, connection pooling, multi-role RBAC, and background PDF generation for admit cards, tabulation sheets, and question banks.
- FocusGuard — Distributed Cross-Platform Attention Platform** [Go 1.22, Docker, WebSockets, Chrome MV3, SQLite] github.com/rofazhasan/FocusGuard
 - Distributed Systems Daemon:** Built a central Go 1.22 service coordinating real-time WebSocket state streaming across desktop, mobile, and browser clients with synchronized aggregate screen-time budgets.
 - Zero-Latency Browser Compilation:** Compiled behavioral policy rules directly into native browser C++ filtering rules via Chrome Manifest V3 declarativeNetRequest, eliminating JavaScript evaluation latency.
 - Local-First Offline Resilience:** Engineered local SQLite policy caching with delta synchronization upon network reconnection; integrated Android VpnService UDP DNS sinkhole intercepting restricted domains with RFC 1035 NXDOMAIN.
- OMRView — In-Browser WebAssembly Computer Vision Engine** [React 19, TypeScript, Vite, OpenCV.js Wasm] github.com/rofazhasan/OMRView
 - WebAssembly CV Pipeline:** Built a client-side document processing engine running compiled OpenCV C++ inside browser WebAssembly threads, eliminating 100% of server-side image processing costs (\$0.00 infrastructure overhead).
 - Geometric Rectification:** Implemented Canny edge detection, Douglas-Peucker polygon approximation, and 4-point homography perspective warp (cv.warpPerspective) to rectify skewed smartphone camera captures in <600ms.
 - Adaptive Intensity Classifier:** Segmented bubble ROIs evaluating pixel intensity histograms against row baseline lighting to accurately grade answer sheets and detect multi-fill collisions.
- Krishi Bondhu AI — Visual Agricultural Diagnostic Assistant** [Flutter, Dart, Riverpod, sqflite, Hardware Camera] github.com/rofazhasan/Krishi_Bondhu_AI
 - Built an edge mobile camera acquisition pipeline standardizing leaf telemetry with local SQLite persistence and bilingual voice playback feedback for offline rural usability.

ACHIEVEMENTS & LEADERSHIP

- Creative Talent Hunt, Regional Champion:** Recognized by the Government of Bangladesh for outstanding academic, mathematical, and computer potential.
- Technical Workshop Organizer:** Organized and taught hands-on peer workshops at RUET on Data Structures, Algorithms, and Object-Oriented System Architecture.
- Competitive Programming & Hackathons:** Active participant in regional programming contests and national software hackathons.

MD. ROFAZ HASAN RAFIU

Generative AI • LLM Systems Engineer

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PROFESSIONAL SUMMARY

Computer Science & Engineering student at RUET specializing in **Generative AI & LLM Systems, Local Small Language Model (SLM) Orchestration**, and structured inference pipelines. Experienced in designing low-temperature prompted evaluation engines (Ollama/Qwen/Llama), Zod schema validation, sub-10ms deterministic error fallbacks, and high-concurrency Next.js 15 / PostgreSQL architectures.

EDUCATION

- Rajshahi University of Engineering & Technology (RUET) — B.Sc. in Computer Science & Engineering**2023 – 2027
 - Admitted through national combined engineering entrance examination (CUET, KUET, RUET) ranking in top merit percentile.
 - Coursework: Algorithms & Data Structures, Object-Oriented Design, Database Systems, Computer Networks, Discrete Math.
- St. Joseph Higher Secondary School — Higher Secondary Certificate (HSC), Science**2022
 - GPA: 5.00 / 5.00

TECHNICAL SKILLS

- Generative AI & LLMs:** Local SLM Deployment (Ollama, Qwen 2.5, Llama 3.2), Prompt Engineering, Structured JSON Extraction, Zod / Schema Enforcement, Context Management, Low-Temperature Sampling, Hallucination Mitigation.
- AI Evaluation & Reliability:** Deterministic Fallback Pipelines, Groundedness Verification, Output Guardrails, Latency Optimization, Edge Inference Strategies, Error Logging & Diagnostic Telemetry.
- Backend & Inference APIs:** Python (FastAPI, Flask), TypeScript, Go (1.22+), Node.js (Next.js 15 App Router), PostgreSQL, Prisma ORM 6.10, Redis (Caching & Rate-Limiting), SQLite, RESTful APIs, WebSockets (Socket.io).
- DevOps & Platforms:** Docker, NGINX, GitHub Actions (CI/CD), Edge Caching (SWR, CDN Cache-Control), Linux / Bash Scripting, Git, Jest.

FEATURED GENAI & LLM PROJECTS

- Digital School AI Mistake Diagnostics Engine** [TypeScript, Next.js 15, Ollama, Qwen 2.5, PostgreSQL, Prisma]Live: rofazacademy.dev
 - Local SLM Inference Pipeline:** Designed and deployed an end-to-end Local SLM inference pipeline utilizing Ollama (qwen2.5:3b / llama3.2:3b) running low-temperature (0.3) system-prompted evaluations over student exam error vectors.
 - Structured Output Extraction:** Enforced strict markdown/JSON schema extraction parsing diagnostic reports into discrete data structures: performance overview, weak topic clusters, and 3-step action recovery plans.
 - Deterministic Fallback Algorithm:** Engineered a sub-10ms fallback routine computing topic error distributions and model explanations when local inference endpoints face latency spikes, ensuring 100% service uptime.
 - Bilingual Transliteration & Neural Conversion:** Integrated phonetic transliteration utilities in TypeScript to standardize English-Bangla academic tokens before model ingestion and dashboard rendering.
- FocusGuard — Policy Compilation & Enforcement Daemon** [Go 1.22, Docker, WebSockets, Chrome MV3, SQLite]github.com/rofazhasan/FocusGuard
 - Rule Compilation Engine:** Built a Go policy distributor compiling dynamic behavioral rules directly into native browser C++ filtering rules via Chrome Manifest V3 declarativeNetRequest for zero-overhead execution.
 - Distributed WebSocket Broadcast:** Maintained real-time synchronization of shared attention budgets across heterogeneous desktop and mobile clients.
 - Local-First Resilience:** Integrated SQLite offline policy caching with vector delta syncing; implemented Android VpnService UDP DNS sinkhole intercepting restricted domains with RFC 1035 NXDOMAIN.
- OMRView — Vision-Assisted Ground-Truth Parser** [React 19, TypeScript, Vite, OpenCV.js Wasm] github.com/rofazhasan/OMRView
 - In-Browser CV Ingestion:** Developed a client-side document rectification pipeline running compiled OpenCV WebAssembly to extract ground-truth answer vectors from camera captures, offloading 100% of compute from the server.
 - Geometric Normalization:** Applied 4-point homography perspective transformations to standardize physical test sheet inputs into structured JSON payloads for downstream automated grading.

ACHIEVEMENTS & LEADERSHIP

- Creative Talent Hunt, Regional Champion:** Awarded by the Government of Bangladesh for demonstrated excellence in mathematics and computer science.
- Technical Workshop Organizer:** Led technical peer workshops at RUET covering Data Structures, Algorithmic Complexity, and Object-Oriented Design.
- Competitive Problem Solving:** Regular participant in university programming contests and regional hackathons.

MD. ROFAZ HASAN RAFIU

Machine Learning • Computer Vision • Applied AI Engineer

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PROFESSIONAL SUMMARY

Machine Learning & Applied Computer Vision Engineer with a strong Computer Science foundation at RUET. Specialized in edge computer vision, in-browser WebAssembly model execution (OpenCV.js, TensorFlow.js), document image preprocessing pipelines, and relational ML data infrastructure. Focused on building zero-infrastructure-cost applied ML solutions running client-side with deterministic fallbacks and scalable backend data pipelines.

EDUCATION

Rajshahi University of Engineering & Technology (RUET) — B.Sc. in Computer Science & Engineering 2023 – 2027

- Admitted through national combined engineering entrance examination (CUET, KUET, RUET) ranking in top merit percentile.
- Relevant Coursework: Algorithms & Data Structures, Linear Algebra, Probability & Statistics, Discrete Mathematics, Database Systems.

St. Joseph Higher Secondary School — Higher Secondary Certificate (HSC), Science 2022

- GPA: 5.00 / 5.00

TECHNICAL SKILLS

Computer Vision & Edge ML: OpenCV.js (WebAssembly), TensorFlow.js (BlazeFace, WebGL), Image Preprocessing (Adaptive Thresholding, Canny Edge Detection, Contour Analysis, 4-Point Homography, Perspective Warp, ROI Segmentation), Tesseract.js (OCR), Pixel Histogram Classification.

Applied AI & Local Inference: Local SLM Deployment (Ollama / Qwen 2.5), Structured Output Extraction, Error Classification Heuristics, Sub-10ms Rule-Based Fallback Routines, Confidence Calibration.

Languages & Frameworks: Python (NumPy, OpenCV basics, FastAPI), TypeScript, Go 1.22, C++, React 19, Next.js 15, Flutter (Dart).

Data & Systems Engineering: PostgreSQL (Prisma ORM, Indexing, Connection Pooling), Redis, SQLite, Docker, Linux/Bash, RESTful APIs, Git, GitHub Actions, Jest.

FEATURED MACHINE LEARNING & VISION SYSTEMS

OMRView — In-Browser Computer Vision Processing Pipeline [TypeScript, React 19, Vite, OpenCV.js Wasm] 2025

Client-side optical mark recognition and document rectification engine | Code: github.com/rofazhasan/OMRView

- Engineered a fully client-side computer vision pipeline utilizing @techstark/opencv-js WebAssembly build, eliminating all backend GPU/server compute costs for document processing.
- Implemented a multistage image transformation sequence: adaptive thresholding, Canny edge detection, Douglas-Peucker polygon approximation, and 4-point homography warp to correct skewed camera captures.
- Designed a bubble fill classification algorithm calculating relative pixel intensity histograms across segmented Regions of Interest (ROIs), achieving resilience across varying lighting and document degradation.
- Architected responsive React 19 UI with real-time visual overlays rendering rectified document contours and instantaneous question mark verification.

Digital School — Vision Monitoring & SLM Mistake Diagnostic Engine [Next.js 15, TensorFlow.js, Ollama, PostgreSQL] 2024 – Present

Production digital education platform with client vision models and local SLM diagnostics | Live: rofazacademy.dev

- Integrated TensorFlow.js BlazeFace face-detection models with client-side WebGL acceleration to monitor examinee presence and orientation during timed online assessments.
- Constructed a local SLM inference service (Ollama / Qwen 2.5) that analyzes student exam error distributions and categorizes mistakes into concrete academic concept weaknesses.
- Engineered deterministic sub-10ms fallback routines that compute accurate topic error metrics if local model inference is delayed or unavailable, maintaining 100% assessment uptime.
- Developed multi-tenant PostgreSQL data models in Prisma with optimized indexing and connection pooling for high-concurrency exam submissions and automated grading.

Krishi Bondhu AI — Crop Disease Mobile Vision Assistant [Flutter, Dart, Riverpod, sqflite, Hardware Camera API] 2024

Offline-first edge mobile image collection and diagnostic application | Code: github.com/rofazhasan/Krishi_Bondhu_AI

- Built an edge mobile image acquisition pipeline capturing and standardizing crop leaf images directly from device camera hardware with real-time framing guides.
- Implemented an offline-first architecture with local SQLite storage, allowing field workers in low-connectivity agricultural areas to record and queue diagnostic data without network dependency.

FocusGuard — Distributed Systems Daemon & WebSocket Agent [Go 1.22, WebSockets, Docker, Chrome MV3] 2025

High-concurrency systems infrastructure for distributed client state tracking | Code: github.com/rofazhasan/FocusGuard

- Engineered a high-performance Go daemon tracking client activity and broadcasting synchronization events over persistent low-latency WebSockets.
- Containerized multi-service environment with Docker Compose and implemented GitHub Actions CI for automated build verification.

ACHIEVEMENTS & LEADERSHIP

Creative Talent Hunt — Regional Champion (Mathematics & Computing) Government of Bangladesh

Selected among top student problem solvers across the division in national competitive talent assessment.

RUET CSE Peer Workshop Instructor & Competitive Programmer 2023 – Present

Conducted technical mentoring sessions on Data Structures, Algorithms, and Object-Oriented Architecture; active participant in university programming contests.